

Mathematical Formulation for NYC Ride Demand Forecasting in 2025

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Abstract

The following is the mathematical formulas in the model for the forecasting of hourly Uber and Lyft ride demand in New York City. Before building the forecasting model, large-scale trip records are cleaned, transformed, and aggregated using DuckDB with SQL and PowerBI, and are integrated with weather and geographic information. The forecasting framework incorporates temporal features, including lagged demand, as well as weather and location-based predictors to capture spatial and temporal variation in ride demand. The project aims to evaluate the predictive performance of different modeling approaches and identify key factors associated with changes in ride demand across New York City.

1 Overview of Final Mathematical Structure for Modeling

- $\hat{D}_{z,c,t}$: predicted hourly demand for pickup zone z , company c , at time t .
- T_t : temporal features, including hour of day, day of week, month, and weekend indicator.
- W_t : weather features, including temperature, rain category, wind category, and snow indicator.
- $D_{z,c,t-k}$: lagged demand, where $k \in \{1, 24, 168\}$ hours.
- $\overline{D}_{z,c,t}^{(w)}$: rolling historical demand over window $w \in \{3, 24, 168\}$ hours.
- $M_{z,c,t}^{(24)}$: maximum observed demand over the previous 24-hour rolling window.
- S_z : pickup-zone feature.
- C_c : ride-hailing company feature.
- H_t : holiday indicator.

- $ZH_{z,t}$: zone-hour interaction feature.

$$\widehat{D}_{z,c,t} = f_{\text{XGB}} \left(T_t, W_t, D_{z,c,t-k}, \overline{D}_{z,c,t}^{(w)}, M_{z,c,t}^{(24)}, S_z, C_c, H_t, ZH_{z,t} \right) + \epsilon_{z,c,t}, \quad (1)$$

2 Target Variable and Lag Features

The target variable is defined as

$Y_{z,c,t}$ = hourly ride demand for pickup zone z , company c , at hour t .

The lag features capture recent, daily, and weekly demand patterns:

$$L_{z,c,t}^{(1)} = Y_{z,c,t-1}$$

$$L_{z,c,t}^{(24)} = Y_{z,c,t-24}$$

$$L_{z,c,t}^{(168)} = Y_{z,c,t-168}, \quad 168 = 7 \times 24.$$

These features represent demand one hour earlier, approximately the same hour on the previous day, and approximately the same hour one week earlier, respectively.

3 Rolling Features

The three-hour rolling mean is defined as

$$MA_{z,c,t}^{(3)} = \frac{1}{3} \sum_{i=1}^3 Y_{z,c,t-i}.$$

The 24-hour rolling mean is defined as

$$MA_{z,c,t}^{(24)} = \frac{1}{24} \sum_{i=1}^{24} Y_{z,c,t-i}.$$

The 168-hour rolling mean is defined as

$$MA_{z,c,t}^{(168)} = \frac{1}{168} \sum_{i=1}^{168} Y_{z,c,t-i}.$$

The maximum demand observed during the previous 24 hours is

$$M_{z,c,t}^{(24)} = \max_{1 \leq i \leq 24} Y_{z,c,t-i}.$$

All rolling features use only observations prior to time t , preventing contemporaneous target leakage.

4 Additional Final Model Features

The holiday indicator is defined as

$$H_t = \begin{cases} 1, & \text{if date } t \text{ is classified as a U.S. holiday,} \\ 0, & \text{otherwise.} \end{cases}$$

To capture zone-specific intraday demand patterns, a categorical zone-hour interaction is defined as

$$ZH_{z,t} = (S_z, h_t),$$

where S_z represents pickup zone and $h_t \in \{0, 1, \dots, 23\}$ represents hour of day.

5 Predictor Vector

The complete predictor vector for the final model is

$$X_{z,c,t} = [T_t, W_t, L_{z,c,t}, R_{z,c,t}, S_z, C_c, H_t, ZH_{z,t}],$$

where the lag vector is

$$L_{z,c,t} = [Y_{z,c,t-1}, Y_{z,c,t-24}, Y_{z,c,t-168}],$$

and the rolling-feature vector is

$$R_{z,c,t} = [MA_{z,c,t}^{(3)}, MA_{z,c,t}^{(24)}, MA_{z,c,t}^{(168)}, M_{z,c,t}^{(24)}].$$

The final XGBoost demand forecast can therefore be expressed as

$$\hat{Y}_{z,c,t} = f_{\text{XGB}}(X_{z,c,t}).$$

6 High Demand Specialization Model Expression

$$\hat{D}_{z,c,t}^{\text{main}} = f_{\text{XGB}}(\mathbf{X}_{z,c,t}). \quad (2)$$

6.1 Residual

$$R_{z,c,t} = D_{z,c,t} - \hat{D}_{z,c,t}^{\text{main}}. \quad (3)$$

6.2 High Demand Correction Model

$$\hat{R}_{z,c,t} = g_{\text{XGB}}(\mathbf{X}_{z,c,t}, \hat{D}_{z,c,t}^{\text{main}}). \quad (4)$$

Finally,

$$\hat{D}_{z,c,t}^{\text{final}} = \hat{D}_{z,c,t}^{\text{main}} + I_{z,c,t}^H \hat{R}_{z,c,t}. \quad (5)$$

Equivalently,

$$I_{z,c,t}^H = \begin{cases} 1, & \hat{D}_{z,c,t}^{\text{main}} \geq \tau, \\ 0, & \hat{D}_{z,c,t}^{\text{main}} < \tau. (\tau = 450 \text{ in model}) \end{cases} \quad (6)$$

7 Demand-Based Sample Weighting

To adjust the contribution of observations from different demand levels during model training, a demand-based sample weighting scheme was applied.

For observation i , the sample weight is defined as

$$w_i = \begin{cases} 1.25, & 0 < Y_i < 50, \\ 3, & 50 \leq Y_i < 500, \\ 4, & Y_i \geq 500, \end{cases}$$

The weighted training objective can be represented as

$$\mathcal{L} = \sum_{i=1}^N w_i (Y_i - \hat{Y}_i)^2.$$

I designed in that way because demand level between 0-5 really have less business impact compared to larger demand level. I give the 0-5 level demand least weight to decrease the emphasis on low-demand prediction errors. For any demand higher than 0-5, I give more than double weight to them for emphasizing high-demand prediction errors.